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Review and critical analysis of The BROCODE model: a novel mathematical model for the Briggs–Rauscher reaction

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Abstract

This project presents a critical analysis of the paper *The BROCODE model: A novel mathematical model for the Briggs-Rauscher reaction* [1], which proposes a mathematical reduction of the Briggs-Rauscher (BR) reaction, an oscillatory chemical system characterized by complex transitions between radical and non-radical processes. We first assess the mathematical consistency of the reduction by performing numerical simulations of an intermediate ODEs system. These simulations address the correctness of the Quasi-Steady State Approximation (QSSA) for low-concentration intermediates, with our results revealing a non negligible sensitivity to parameter selection. The analysis is further extended to a stochastic framework. By systematically varying the reaction volume, we demonstrate that deterministic and stochastic descriptions converge only in the large-volume (i.e. large amount of reactants) limit. Finally, we investigate the BR reaction as an antioxidant assay by simulating the introduction of an antioxidant species (namely, ferulic acid). Sustained inhibition of oscillations is observed only when the simultaneous scavenging of both hydroperoxyl and iodine dioxide radicals is taken into account. These findings highlight the chemical complexity of the system and emphasize the limitations of reduced models.

1. Introduction

The authors derive a new approximate mathematical model for the Briggs–Rauscher (BR) reaction (see [1]), an oscillatory chemical system characterised by periodic fluctuations in the concentrations of several reacting species. Although detailed mechanistic models of the reaction already exist, they are often analytically intractable due to their complexity, involving up to fifteen coupled differential equations.

In contrast, the model analysed in this paper provides a substantial simplification, reducing the system to just three differential equations governing the concentrations of iodous acid, iodide, and molecular io-

dine. This reduction aims to retain the essential dynamical features of the reaction while improving mathematical transparency and analytical accessibility.

The Briggs–Rauscher reaction is particularly attractive as a model system because it is relatively easy to observe experimentally. In the laboratory, the reaction exhibits striking oscillations between yellow and dark blue colour states, corresponding to changes in iodine and triiodide concentrations in the presence of starch. While the BR reaction shares qualitative similarities with other oscillatory systems, such as the Bray–Liebhafsky reaction, it differs in important respects. In particular, the maximum iodide concentration attained in the BR reaction is significantly higher,

and the reaction is readily observable under standard laboratory conditions.

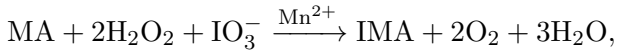
1.1. Reaction Mechanism

The Briggs–Rauscher reaction exhibits oscillatory behaviour arising from the competition between two interconnected oxidation–reduction processes, that alternate over time:

- **Radical process (blue phase):** When the iodide (I^-) concentration is low, the reaction proceeds through a radical-dominated mechanism. Manganese ions (Mn^{2+}) catalyse the formation of reactive iodine species, including hypoiodous acid (HOI). This leads to an increase in triiodide (I_3^-). Radical species such as $IO_2^\bullet$ and $HO_2^\bullet$ further influence the reaction kinetics. In the presence of starch, this results in a dark blue colour.
- **Non-radical process (yellow phase):** As iodide accumulates, it suppresses the radical pathway and the reaction shifts to a non-radical mechanism. Iodine is then consumed primarily through its reaction with malonic acid (MA), reducing the triiodide concentration and causing the solution to return to an amber or yellow colour. Typically, this phase lasts less than the blue one.

Oscillations arise because the reaction between iodine and malonic acid proceeds more slowly than the other reaction steps. This separation of timescales allows intermediate species to accumulate and then be rapidly consumed, producing periodic changes in concentration.

The overall chemistry is commonly summarised by the net reaction



which describes the halogenation of malonic acid (MA) in the presence of hydrogen peroxide, iodate, and a manganese catalyst to form iodomalonic acid (IMA), oxygen, and water.

Although alternative formulations of the Briggs–Rauscher reaction exist, this variant is the most widely studied.

1.2. Biological Application

Beyond its theoretical interest, the Briggs–Rauscher reaction has also been proposed as a practical assay for measuring antioxidant activity in foodstuffs [2] [3]. In this application, a sample is introduced at the onset of oscillations, temporarily suppressing them for a duration proportional to its antioxidant capacity. Compared with existing methods, this assay is rapid, simple, and operates at a pH comparable to that of the

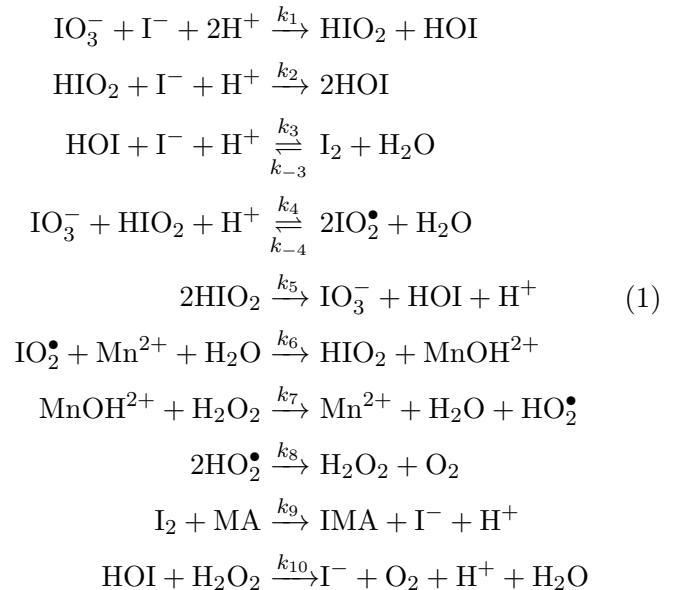
human stomach. Interestingly, while many polyphenolic compounds halt oscillations almost immediately, salicylic acid (a simple monophenolic compound) has been observed to suppress oscillations more gradually, highlighting subtle differences in antioxidant behaviour within this chemically rich system [4].

2. Starting model

In the analyzed paper the authors start with the De Kepper and Epstein model [5] which consists of ten steps of which two are reversible. The overall kinetic behaviour requires the consideration of 15 independent species.

Formula	Common Name
IO_3^-	Iodate ion
I^-	Iodide ion
I_2	Molecular iodine
HOI	Hypoiodous acid
HIO_2	Iodous acid
$IO_2^\bullet$	Iodine dioxide radical
$HO_2^\bullet$	Hydroperoxyl radical
Mn^{2+}	Manganese(II) ion (catalyst)
$MnOH^{2+}$	Manganese intermediate complex
H_2O_2	Hydrogen peroxide
MA	Malonic acid
IMA	Iodomalonic acid
O_2	Oxygen
H^+	Hydrogen ion
H_2O	Water

Table 1: Key chemical species in the Briggs–Rauscher reaction.



Each k_i denotes a rate constant, that whenever possible has been experimentally determined.

Constant	Value	Units
k_1	1.43×10^3	$\text{M}^{-3} \text{s}^{-1}$
k_2	2.0×10^{10}	$\text{M}^{-2} \text{s}^{-1}$
k_3	3.1×10^{12}	$\text{M}^{-2} \text{s}^{-1}$
k_{-3}	2.2	s^{-1}
k_4	7.3×10^3	$\text{M}^{-2} \text{s}^{-1}$
k_{-4}	1.7×10^7	$\text{M}^{-1} \text{s}^{-1}$
k_5	6.0×10^5	$\text{M}^{-1} \text{s}^{-1}$
k_6	1.0×10^4	$\text{M}^{-1} \text{s}^{-1}$
k_7	3.2×10^4	$\text{M}^{-1} \text{s}^{-1}$
k_8	7.5×10^5	$\text{M}^{-1} \text{s}^{-1}$
k_9	40	$\text{M}^{-1} \text{s}^{-1}$
k_{10}	37	$\text{M}^{-1} \text{s}^{-1}$

Table 2: Rate constants used in the Briggs-Rauscher reaction model of Kepper and Epstein

3. Reduced model

To reduce complexity, the authors make several simplifying assumptions.

First, they treat the concentration of water as constant, reasoning that its large excess relative to other reactants makes any variation negligible.

Second, they remove oxygen and iodomalonic acid from the system because these species appear only as products, and therefore do not influence the rates of other reactions¹, and similarly, the hydroperoxyl radical is shown to be decoupled from the rest of the dynamics, allowing it to be ignored.

Third, applying a conservation argument they eliminate manganese hydroxide. In fact, since only Mn^{2+} and MnOH^{2+} are manganese containing species and the first of these is the only present at the start $t = 0$, it holds that

$$[\text{MnOH}^{2+}] = [\text{Mn}^{2+}]_0 - [\text{Mn}^{2+}].$$

Replacing $[\text{MnOH}^{2+}]$ by this formula, its equation can be eliminated.

Fourth, several species, including malonic acid, hydrogen peroxide, iodate and hydrogen ions, are considered “pool” chemicals, whose concentrations remain essentially unchanged over the timescale of the reaction.

Fifth, they assume the manganese concentration as effectively constant, justified by experimental and numerical evidence of minimal change.

Finally, low-concentration intermediates, specifically iodine dioxide (IO_2) and hypoiodous acid (HOI), are treated using a quasi-steady state approximation, assuming their rates of change are negligible compared with other species. These simplifications reduce the system to a three-dimensional ODE model (the

¹From a modelling point of view, this has the drawbacks that the trajectories diverge to $+\infty$ as $t \rightarrow +\infty$. This behaviour can be traced back to the approximations underlying the reduction of the model.

BROCODE model) that reproduces the main oscillatory dynamics.

$$\begin{aligned} \frac{dU}{dt} &= k_1 a h^2 V - k_2 h U V - 2k_5 U^2 + \frac{1}{2} k_6 c m \cdot S(U) \\ \frac{dV}{dt} &= -k_1 a h^2 V - k_2 h U V + k_{-3} c Z + k_9 q Z + \\ &\quad (k_{10} b - k_3 h V) \cdot W(U, V, Z) \\ \frac{dZ}{dt} &= k_3 h V \cdot W(U, V, Z) - k_{-3} c Z - k_9 q Z \end{aligned}$$

where $U = [\text{HIO}_2]$, $V = [\text{I}^-]$, $Z = [\text{I}_2]$,

$$S(U) = \frac{1}{4k_{-4} a h} (-k_6 c m + \sqrt{(k_6 c m)^2 + 16k_4 k_{-4} a h c U})$$

and

$$W(U, V, Z) = \frac{k_1 a h^2 V + 2k_2 h U V + k_{-3} c Z + k_5 U^2}{k_3 h V + k_{10} b}$$

However, the assumptions rely on restrictive approximations regarding species concentrations and timescales, which could neglect subtle interactions present in the full reaction network.

4. Numerical simulation

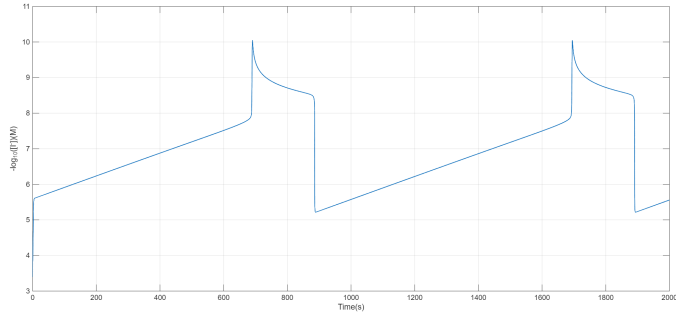
To verify the well-posedness of the quasi-steady state approximation and the assumption of constant manganese concentration, we performed a numerical simulation of the intermediate system. This system consists of six ODEs derived after applying the first four simplifications but prior to applying the QSSA and constant manganese assumptions:

$$\begin{aligned} \frac{d[\text{I}^-]}{dt} &= -k_1 a h^2 [\text{I}^-] - k_2 h [\text{HIO}_2] [\text{I}^-] - k_3 [\text{I}^-] [\text{HOI}] \\ &\quad + k_{-3} c [\text{I}_2] + k_9 q [\text{I}_2] + k_{10} b [\text{HOI}] \\ \frac{d[\text{HOI}]}{dt} &= k_1 a h^2 [\text{I}^-] + 2k_2 h [\text{HIO}_2] [\text{I}^-] - k_3 h [\text{I}^-] [\text{HOI}] \\ &\quad + k_{-3} c [\text{I}_2] + k_5 [\text{HIO}_2]^2 - k_{10} b [\text{HOI}] \\ \frac{d[\text{HIO}_2]}{dt} &= k_1 a h^2 [\text{I}^-] - k_2 h [\text{HIO}_2] [\text{I}^-] - k_4 h [\text{IO}_3^-] [\text{HIO}_2] \\ &\quad + k_{-4} c [\text{IO}_2^\bullet]^2 [\text{Mn}^{2+}] \\ \frac{d[\text{I}_2]}{dt} &= k_3 h [\text{I}^-] [\text{HOI}] - k_{-3} [\text{I}_2] - k_9 q [\text{I}_2] \\ \frac{d[\text{IO}_2^\bullet]}{dt} &= 2k_4 a h [\text{HIO}_2] - 2k_{-4} c [\text{IO}_2^\bullet]^2 - k_6 c [\text{Mn}^{2+}] [\text{IO}_2^\bullet] \\ \frac{d[\text{Mn}^{2+}]}{dt} &= -k_6 c [\text{Mn}^{2+}] [\text{IO}_2^\bullet] + k_7 b ([\text{Mn}]_0 - [\text{Mn}^{2+}]) \end{aligned}$$

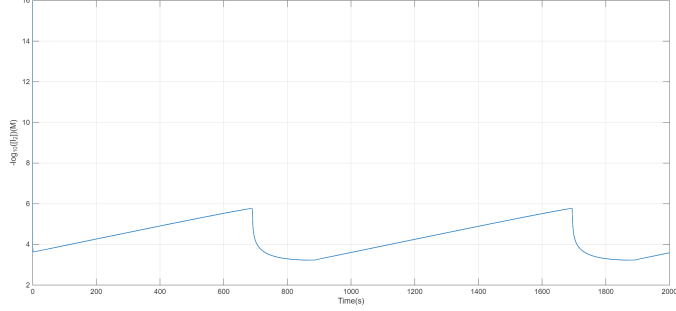
Here, the parameters represent the pool chemicals $a = [\text{IO}_3^-]$, $b = [\text{H}_2\text{O}_2]$, $c = [\text{H}_2\text{O}]$, $h = [\text{H}^+]$ and $q = [\text{MA}]$.

The simulation utilized the *ode15s* solver with relative and absolute tolerances set to 10^{-10} and 10^{-16} , respectively. The initial conditions and rate constants used are detailed in Tables 4 and 3.

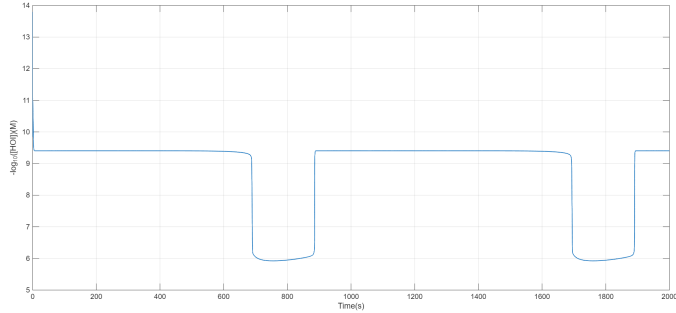
The shape of the curves reveals the oscillatory behaviour of the reaction and in particular highlights its



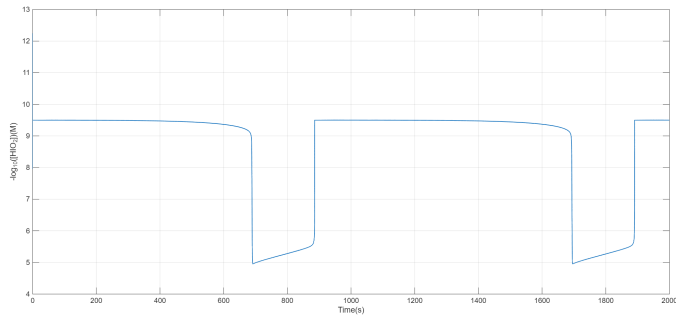
(a) $-\log_{10}$ of the molar concentration of I^- over time



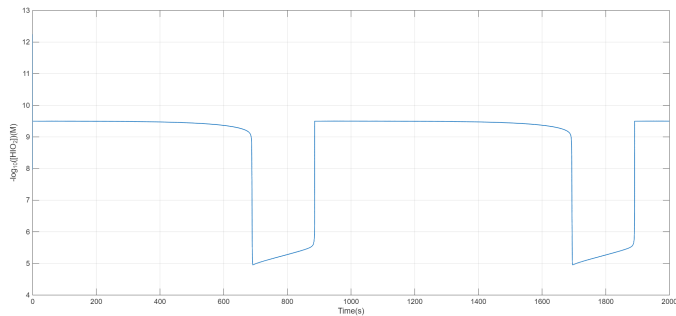
(b) $-\log_{10}$ of the molar concentration of I_2 over time



(c) $-\log_{10}$ of the molar concentration of HOI over time



(d) $-\log_{10}$ of the molar concentration of HIO_2 over time



(e) $-\log_{10}$ of the molar concentration of $IO_2^\bullet$ over time

Figure 1

Constant	Value	Units
k_1	5×10^3	$M^{-3} s^{-1}$
k_2	2×10^{10}	$M^{-2} s^{-1}$
k_3	3.1×10^{12}	$M^{-2} s^{-1}$
k_{-3}	0.0130	s^{-1}
k_4	7.35×10^3	$M^{-2} s^{-1}$
k_{-4}	8×10^5	$M^{-1} s^{-1}$
k_5	1×10^5	$M^{-1} s^{-1}$
k_6	180.2×10^4	$M^{-1} s^{-1}$
k_7	3.2×10^4	$M^{-1} s^{-1}$
k_8	7.5×10^5	$M^{-1} s^{-1}$
k_9	12.5	$M^{-1} s^{-1}$
k_{10}	37	$M^{-1} s^{-1}$

Table 3: Rate constants used in the numerical simulation

distinct phases, which are clearly visible in the plot for iodide (I^-) 3a. We can divide the process in two phases: one is slow and the other is fast. The slow phase is highlighted by long periods (e.g. from $t = 0$ to 700) in which the curves slope gently. This represents a slow consumption phase where the system gradually drifts away from the initial condition. Suddenly, the system reaches a tipping point, resulting in near-vertical jumps in concentration. This indicates a rapid transition where chemical changes occur almost instantaneously. We can also observe that the iodide (I^-) acts as the master control switch for the entire system. When its concentration is high (depicted as low values on the negative logarithmic scale), it acts as an inhibitor, preventing the rapid production of the other species. Once iodide is consumed below a critical level (at approximately $t \approx 700$), this inhibition is lifted. This triggers a sudden explosion in the production of chemicals like $HO_2^\bullet$ and $IO_2^\bullet$, causing the visible jumps in their respective plots 3b and 3c.

The numerical data in the original paper provided a strong justification for the final two simplification steps, but there is a caveat. While in the original paper it was shown 2b that some reactants such as $IO_2^\bullet$ and HOI are $O(10^{-7})$ (so one or more orders of magni-

Variable	Initial Value
a	0.0225 M
b	0.33 M
h	0.056 M
q	0.0015 M
c	55.5 M
$[I^-]$	0.0004 M
$[HOI]$	0 M
$[HIO_2]$	0 M
$[I_2]$	0 M
$[IO_2^\bullet]$	10^{-5} M
$[Mn^{2+}]$	0.004M

Table 4: Initial conditions used in the numerical simulation

tude less than the other reactants) in our simulations this hypothesis was still valid with the set of parameters given by [5] and present in Table 2 (see 2c) but appears to be weakened by the assumption of the new rates (2d). Hence, it seems that the different choice of the parameters strongly weakens the hypothesis underlying the reduction of the old model and the formulation of the new one. Another evidence detected by the comparison of the two simulations is that the change of rate constants relaxes the oscillating time. Indeed, while with the rate constants given by [5] and present in Table 2, we notice up to 5 oscillations in the time interval $[0, 1500]$, using the parameters of BROCODE in the same interval we obtained only one oscillation.

Finally, it is noticeable from the Figure 2a that the concentration of manganese exhibits minimal change over time, validating the assumption that it can be treated as a constant in the reduced model.

Using the same initial conditions and rate constants (Tables 4 and 3), we analyzed the system's limit cycle 3d which connects the behavior of the two "slow" variables, iodine (I_2) and iodide (I^-), into a single phase-space curve. After an initial transient phase determined by the initial conditions, the system settles onto the limit cycle, evolving from left to right. According to the time evolution of $[I^-]$, the system follows the slow branch at the bottom until the first tipping point. At this point, the fast dynamics takes over, leading to a rapid decrease in $[I^-]$ (or, equivalently, to an increase in $-\log_{10}([I^-])$) until the second tipping point is reached. The dynamics then switches back to the slow regime (the yellow branch). Both I^- and I_2 increase until the third tipping point, where another transition to fast dynamics occurs. Finally, the system reaches the fourth and last tipping point and continues along the cycle.

5. Extensions

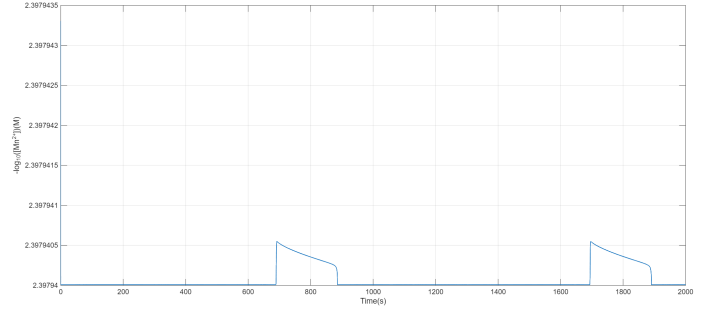
5.1. Stochastic simulation

5.1.1 Introduction

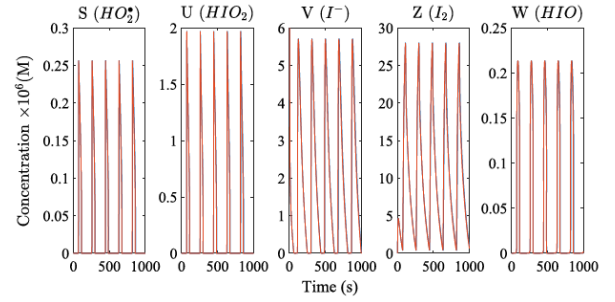
We investigate the system using stochastic simulations as a complementary approach to the deterministic ODE analysis. Beyond providing a consistency check for the continuum approximation underlying the ODE model, this framework allows us to explore dynamical regimes and fluctuations that cannot be captured within a purely mean-field description.

5.1.2 Model and constant conversion

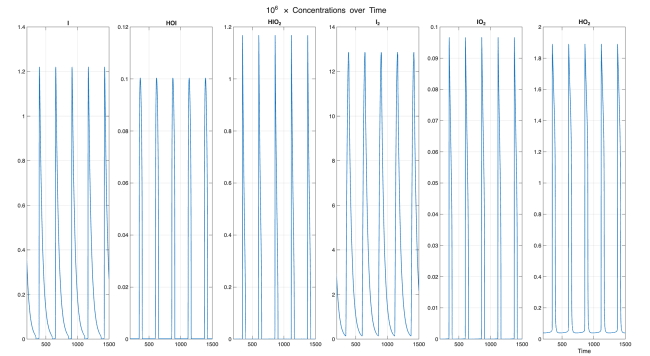
Following the assumptions adopted in the reference paper[1], the concentrations $[H_2O]$, $[IO_3^-]$, $[H_2O_2]$, $[H^+]$, and $[MA]$ are treated as constant. From a stochastic modeling perspective, this implies that the cor-



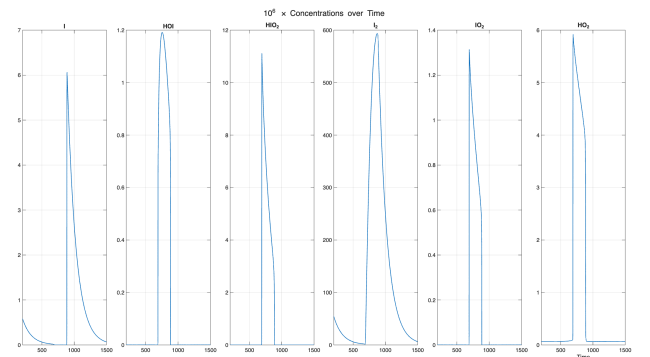
(a) $-\log_{10}$ of the molar concentration of Mn^{2+} over time



(b) Original figure from [1]. The rate constant k_{-4} is not specified in the original work and is assumed not to significantly affect the dynamics. Note that there is a small typo in the paper, and $HO_2^\bullet$ appears instead of $IO_2^\bullet$, on the left.

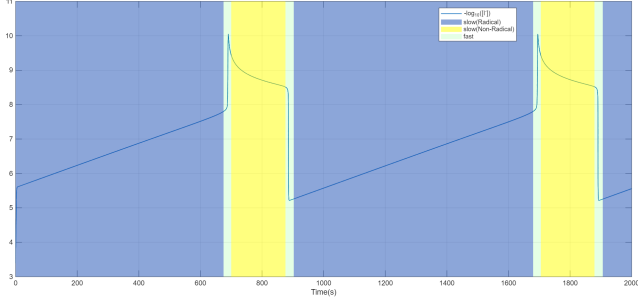


(c) Deterministic simulation using parameters from [5] with $k_{-4} = 10^7 \text{ M}^{-2}\text{s}^{-1}$.

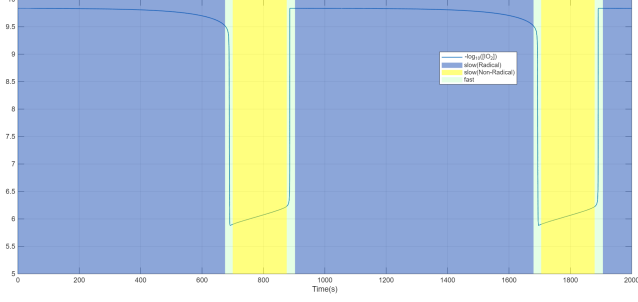


(d) Deterministic ODE simulation using the same parameter set of BROCODE.

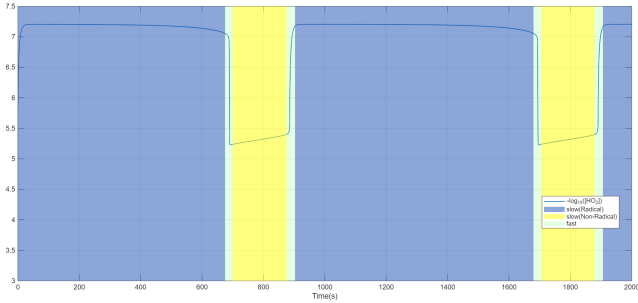
Figure 2: Comparison between the original results reported in [1] and the corresponding stochastic and deterministic simulations obtained in this work.



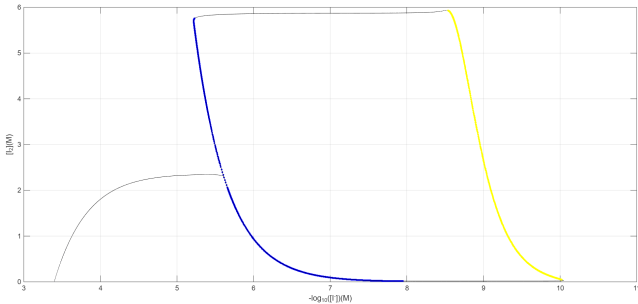
(a) $-\log_{10}$ of the molar concentrations of I^- over time



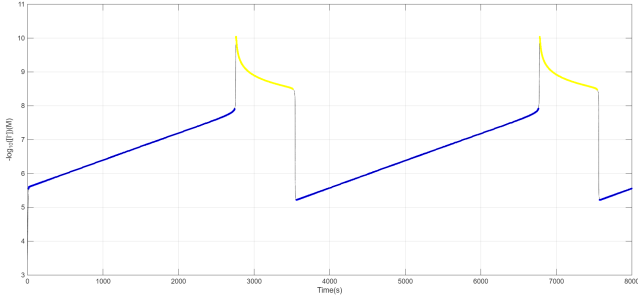
(b) $-\log_{10}$ of the molar concentrations of $IO_2^\bullet$ over time



(c) $-\log_{10}$ of the molar concentrations of $HO_2^\bullet$ over time



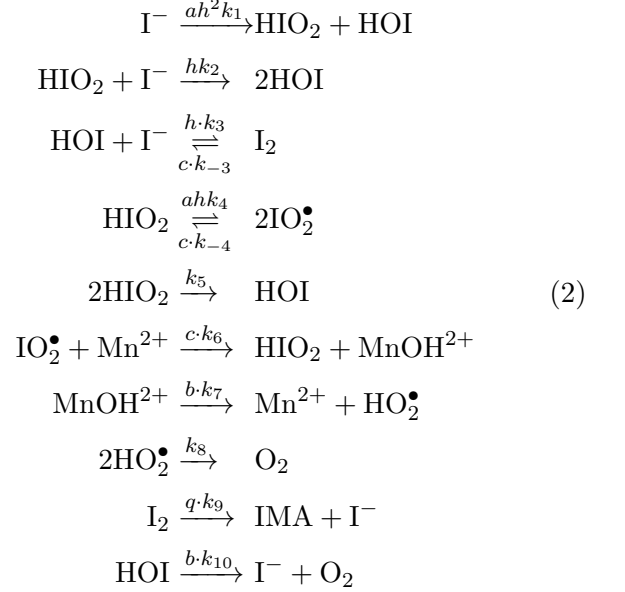
(d) Evolution in the iodide/iodine phase space



(e) $-\log_{10}$ of the molar concentration of I^- over time

Figure 3

responding species can be removed from the explicit reaction network and their effect incorporated into rescaled reaction rates. The resulting effective system reads:



The species assumed to be constant are therefore no longer explicitly present in the system. Their contribution is accounted for through rescaling factors $a = 0.01$, $b = 0.33$, $h = 0.056$, $q = 0.0015$, and $c = 55.5$, corresponding respectively to the concentrations of $[IO_3^-]$, $[H_2O_2]$, $[H^+]$, $[MA]$, and $[H_2O]$. As a result, we retain the same number of reactions as in the original model, but with a reduced number of dynamic species.

The next step consists in specifying meaningful initial conditions. To this end, we refer to the deterministic ODE system. Since stochastic simulations require integer-valued molecule counts, molar concentrations are converted into absolute numbers of molecules. Throughout the manuscript, N_A denotes Avogadro's number. Based on the typical orders of magnitude involved, we select an arbitrary reaction volume:²

²Some species in the deterministic ODE model reach concentrations as low as $10^{-10} M$. In the stochastic setting, concentrations are effectively rescaled by a factor of 10^8 through the choice of the system volume, implying that such low-abundance species fall below the resolution of the discrete description. The choice of the reaction volume therefore represents a trade-off between resolving rare species and maintaining a feasible computational cost. Larger volumes would improve the resolution of low-concentration species, but at the price of a dramatic increase in total reaction propensities and system stiffness, leading to prohibitively expensive stochastic simulations. As a consequence, the stochastic dynamics cannot be expected to be quantitatively equivalent to the deterministic ODE solution, either in terms of absolute magnitudes or in the detailed structure of the trajectories. Nevertheless, despite this intrinsic limitation and the loss of several orders of magnitude, the stochastic simulations still exhibit clear oscillatory behavior, indicating that the qualitative dynamical features of the system are robust with respect to the modeling framework.

$$V = \frac{10^8}{N_A} \text{ l} \approx 1.6605 \times 10^{-16} \text{ l}$$

This choice directly affects the stochastic rate constants, which must be rescaled according to the formula:

$$c_j = \frac{k_j}{(V \times N_A)^{\text{order}_j - 1}} \prod_{i=1}^8 v_{ji}^-!$$

Index	Species	Initial concentration [M]	Initial number of molecules
1	I ⁻	4×10^{-4}	4×10^4
2	HOI	0	0
3	HIO ₂	0	0
4	I ₂	0	0
5	IO ₂ [•]	10^{-5}	10^3
6	Mn ²⁺	0.004	4×10^5
7	HO ₂ [•]	10^{-3}	10^5
8	MnOH ²⁺	0	0

Table 5: Initial concentrations and corresponding molecule numbers. Unlike the other species, the initial abundance of $HO_2^{\bullet-}$ was not taken from the reference paper but was set manually for numerical reasons.

Reaction	Deterministic rate	Stochastic rate
1	5×10^3	0.3528
2	2×10^{10}	11.20
3	3.1×10^{12}	1736
4	7.35×10^3	9.261
5	1×10^5	0.002
6	$c \cdot 180.2$	1.00011×10^{-4}
7	3.2×10^4	10560
8	7.5×10^5	0.015
9	12.5	0.01875
10	37	12.21
11	$c \cdot 0.013$	0.7215
12	$c \cdot 8 \times 10^5$	0.888

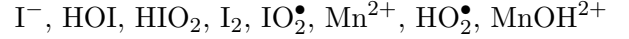
Table 6: Deterministic and stochastic reaction rates.

Reactions 11 and 12 correspond to the reverse processes associated with the rates k_{-3} and k_{-4} , respectively.

5.1.3 Simulations

The dynamics were simulated using an implementation of the Direct Method exploiting a dependency graph (*Enhanced Direct Method*). The resulting time series were subsampled every 10^{-3} time units. Like in the deterministic ODE system, a subtle issue arises in the stochastic formulation: the species O₂ and IMA appear only as products and therefore accumulate indefinitely, diverging as time tends to infinity³. In the

deterministic setting, the corresponding equations are not required to compute the dynamics of the remaining species, and this issue does not arise. In the stochastic framework we will proceed similarly. Columns of the stoichiometric matrices are ordered as



Eventually, we have the following matrices:

$$v_{\text{BR}}^- = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \end{pmatrix}$$

$$v_{\text{BR}}^+ = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

After a transient phase, the system clearly reaches an oscillatory regime 4a. As expected, I₂ exhibits the largest molecule count, while the remaining species populate distinct orders of magnitude. Interpretation of the stochastic results requires caution. Some species in the deterministic system reach concentrations around 10^{-10} M, below the resolution allowed by our chosen reaction volume (effectively scaled by 10^8). Consequently, low-abundance species often reach zero in the stochastic simulations and appear noisier than in the ODE results (see Figure 4b). This issue is particularly evident on a logarithmic scale (Figure 4c).

More generally, one should not expect a one-to-one quantitative correspondence between the stochastic and deterministic descriptions, neither for individual species nor for the relative ratios between concentrations. This discrepancy is an intrinsic consequence of the discrete nature of the stochastic model and of

From a modeling perspective, this behavior can be traced back to the approximations adopted so far, such as assuming the concentration of MA to remain constant despite its involvement in the production of IMA.

³As a consequence, their molecule counts may become very large, posing numerical challenges for long-time simulations.

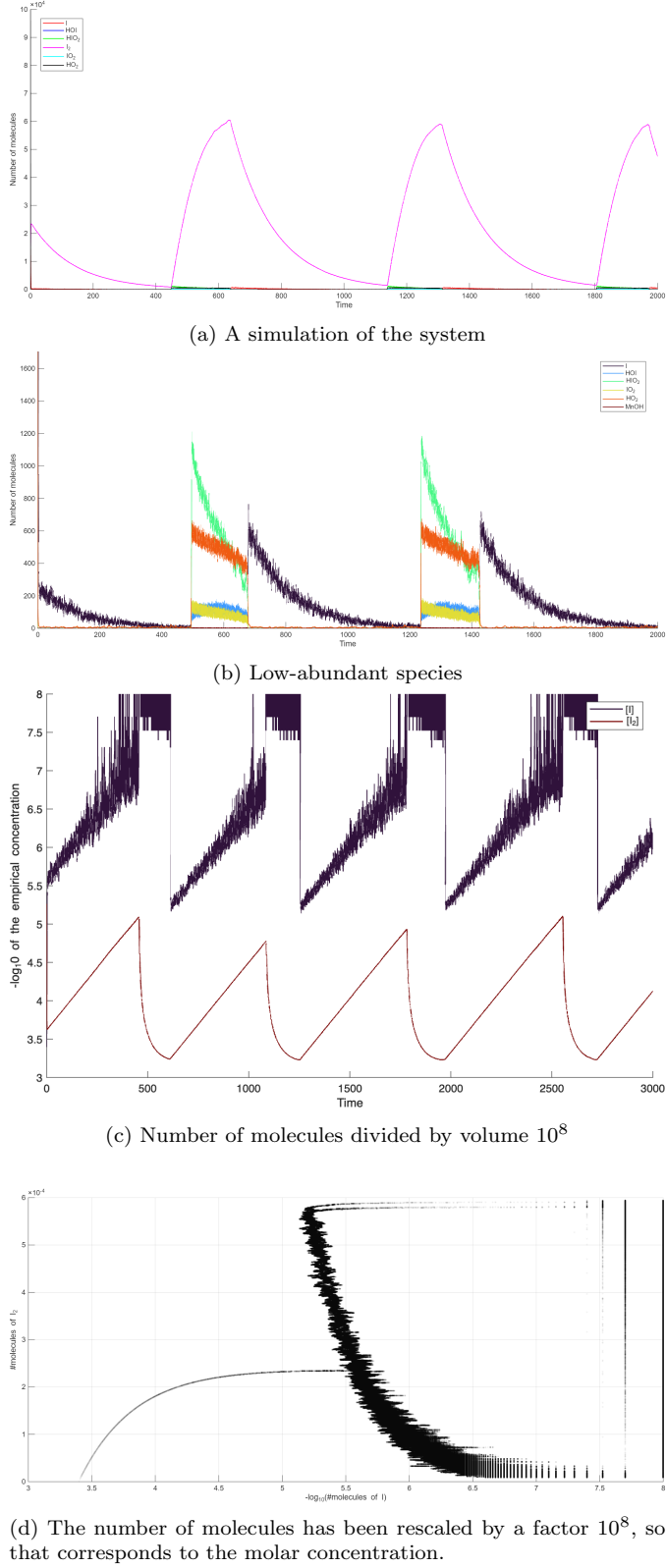


Figure 4

the loss of several orders of magnitude associated with the chosen system volume. Nevertheless, despite these limitations, the stochastic simulations are sufficient to reveal a reasonably regular oscillatory behavior over time. In particular, for highly abundant species such as I_2 , the stochastic concentration closely resembles its deterministic counterpart. The oscillations are irregular, (see 4d) as expected for a stochastic system and, due to the chosen system volume, cannot be expected to quantitatively correspond to the ODE oscillations when the concentration of I^- is around 10^{-8} i.e. in the right-hand part of the plot.

The system was simulated up to $T_{\max} = 10^5$ s. No bistability phenomena, such as those observed in stochastic Lotka–Volterra systems, were detected. The simulation took approximately 1745 seconds (≈ 30 minutes) to complete, highlighting the computational cost and motivating the exploration of more efficient simulation algorithms (if any).

5.1.4 Benchmarks

This section compares the performance of several exact stochastic simulation algorithms applied to the reduced reaction network, with the goal of facilitating future computational studies. For each method, execution times are reported for increasing final simulation times $T_{\max}$, while keeping the sampling interval fixed at $dT = 10^{-2}$. Whenever supported, simulations were executed in parallel. All benchmarks were performed on a MacBook Air (2025) equipped with an Apple M4 processor (ARM64 architecture), running macOS. Reported execution times correspond to wall-clock times and were averaged over 100 independent simulations. The *Next Reaction Method* was not included in the benchmark comparison, as it proved unsuitable for the present system due to numerical stability issues arising from vanishing propensities. A detailed discussion of these limitations, together with illustrative simulations, is provided in the appendix.

Alg. $\backslash T_{\max}$ [s]	10^2	2×10^2	5×10^2	10^3
EDM	2.4416	3.5238	9.6201	30.0018
DM	6.1134	9.2916	24.3305	74.7792
RSSA*	9.6303	13.8283	31.8367	125.0828
FRM	6.1484	8.9848	25.0621	77.4519

Table 7: Execution times for different methods ($dT = 10^{-2}$), averaged over 100 simulations. * indicates a dependency graph has been exploited. For RSSA, δ was set to 0.1.

As expected, all methods exhibit a non linear increase in execution time with respect to $T_{\max}$, with EDM consistently outperforming the other algorithms. The choice of the parameter δ in RSSA affects the algorithm performance (see Figure 6a). Among the tested values, $\delta = 0.1$ provides one of the best trade-offs in terms of execution time. For each value of δ , 20 simulations were performed, and the reported execution

times correspond to the averages over these runs, for $T_{max} = 500$ s. In 6.4 we spend few word on RSSA applied in this context.

5.1.5 Increasing the volume

When the system volume is increased (by approximately two orders of magnitude), the stochastic dynamics of the less abundant species become less and less noisy and the resulting trajectories increasingly resemble those obtained from the deterministic ODE model. This improvement comes at the cost of a substantially higher computational burden: for instance, simulations with

$$V = \frac{10^{10}}{N_A} l \approx 1.6605 \times 10^{-14} l$$

require on the order of an hour of computation when $T_{max} = 2000$ s and $dT = 0.01$.

Moreover, the molar concentrations inferred from the stochastic simulations show progressively better agreement with the corresponding deterministic solutions, reproducing qualitatively similar dynamical behavior.⁴ These observations 6d 6c support the validity of the ODE framework as an effective approximation of the system dynamics in the large-volume limit, especially due to the advantage in runtime.

Finally, it is worth noting that the characteristic time scale of the stochastic dynamics appears to increase as the system volume is increased (about four complete oscillations in 8c, two in 4b, one in 6b). Although a definitive explanation lies beyond the scope of the present work, this behavior may be related to the intrinsic rescaling of reaction rates in the stochastic formulation. In particular, while smaller volumes are often associated with increased collision frequencies when the number of molecules is held fixed, in our setting molecular abundances are rescaled so as to pre-

⁴As said above, no phenomena such as bistability, or extinction of a species, as observed for instance in stochastic Lotka-Volterra systems, were detected.

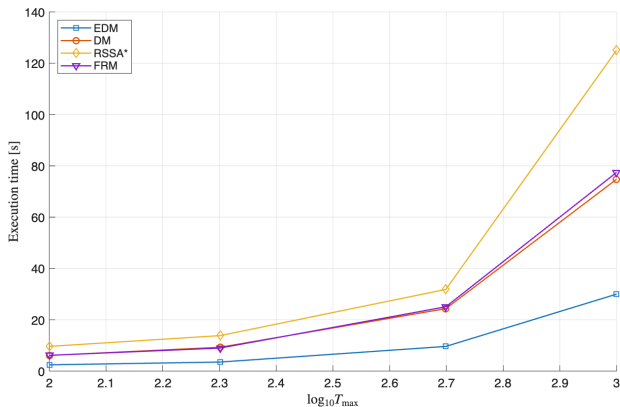
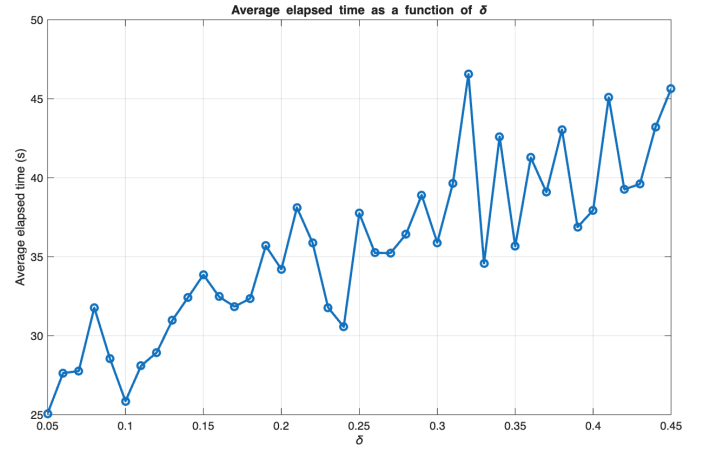
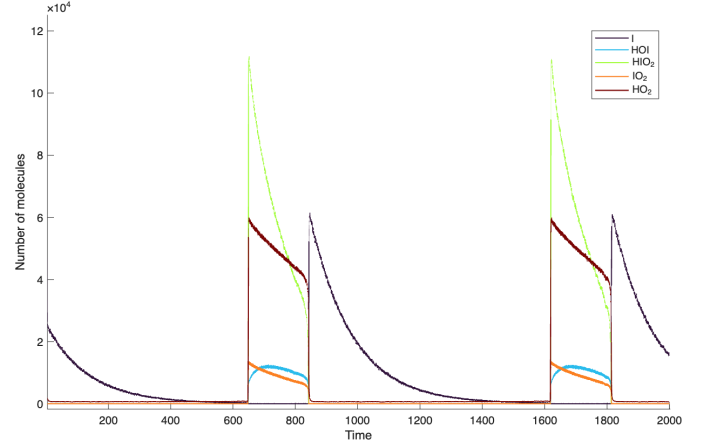


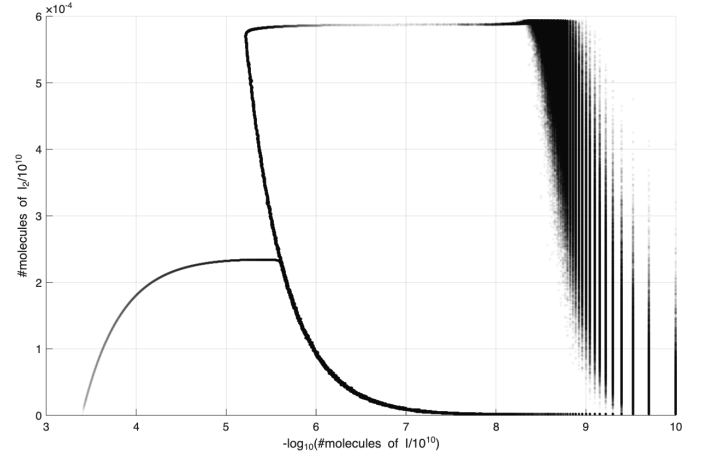
Figure 5: Comparative graph for the time complexities.



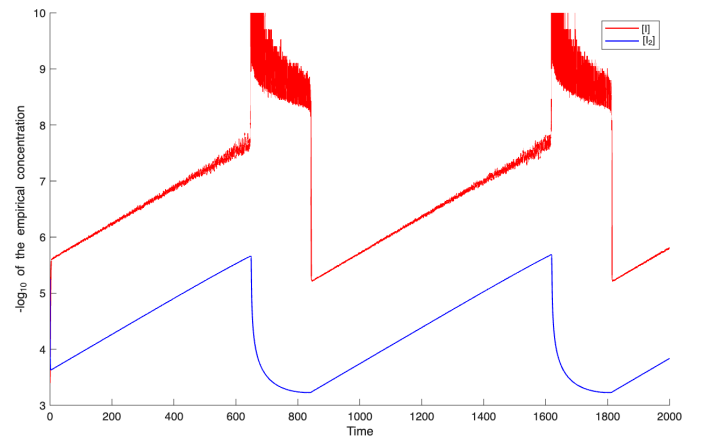
(a) Average time for different values of δ



(b) Low abundance chemicals when $V = 10^{10}/N_A$ l.



(c) The oscillations become more regular, and we approach more and more the limit cycle of the ODE. The number of molecules has been divided by 10^{10} so that what it's plotted is just the concentration.



(d) The number of molecules has been rescaled by a factor 10^{10} , so that it corresponds to the molar concentration.

serve concentration ratios ⁵. As a result, the observed acceleration of the dynamics should be interpreted as an emergent effect of the stochastic scaling rather than a simple kinetic consequence.

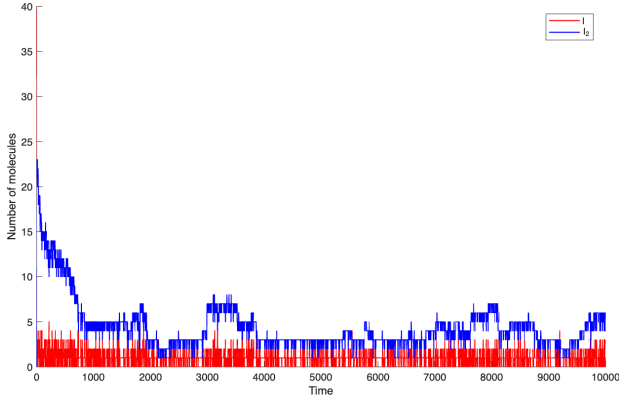


Figure 7: $V = 10^5/N_A \text{ l}$

5.1.6 Decreasing the volume

It is remarkable that the initial conditions reported in Table 5 still remain meaningful when $V = 10^5/N_A \text{ l}$. The dynamics becomes very noisy for low-abundance reactants, yet the system is still capable of reproducing oscillations for some species, such as I_2 . Moreover, $[I^-]$ never exceeds 10^{-5} M in the ODEs solution, so one might expect it to remain effectively zero when scaling all abundances by 10^5 , but this is clearly not the case (see Figure 7). This highlights that, for low-abundance species, the deterministic and stochastic frameworks are no longer equivalent. For completeness, we also report representative trajectories for the cases $V = 10^6/N_A \text{ l}$ and $V = 10^7/N_A \text{ l}$ (see Figure 8), so as to provide the reader with an indication of the order of magnitude required for the stochastic dynamics to safely approximate the behavior of the reactants within the deterministic ODE framework.

⁵One should be careful when interpreting “ratio” in this context, since expressions like “0.33 molecules” are not physically meaningful. This fact may partly explain why the observed dynamics are faster than in the ODEs approach, as the concentration ratios are not strictly preserved beyond the initial conditions.

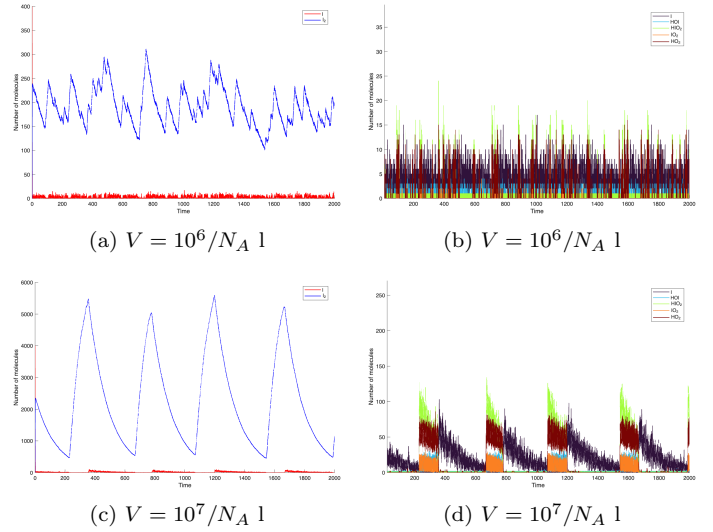


Figure 8: Stochastic dynamics for different reaction volumes. Top row: $V = 10^6/N_A \text{ l}$. Bottom row: $V = 10^7/N_A \text{ l}$.

From a biological point of view, each of the studied volumes could correspond to meaningful cellular or subcellular structures.

Volume [L]	Typical biological scale
$10^{-14} - 10^{-15}$	Large bacteria, lymphocytes, small eukaryotic cells
$10^{-16} - 10^{-17}$	Vesicles, cytoplasmic microcompartments, endosomes, mitochondria
$10^{-18} - 10^{-19}$	Small molecular compartments, large proteins, or macromolecular complexes

Table 8: Typical biological objects corresponding to different cellular volumes.

5.1.7 Few words on the full BR system

Reproducing the dynamics of the full BR system requires some caution. On the one hand, the stochastic rates of the reactions of order 3 and 4 become increasingly negligible as the volume increases, eventually becoming virtually zero. In practice, this means that some reactions are never fired. On the other hand, keeping a small scale factor results in noisy trajectories that, even in the reduced model, are different from the deterministic ones. For completeness, we report one of the trajectories for $V = 10^6/N_A \text{ l}$ (see Figure 9).

5.2. Antioxidant Activity

The Briggs–Rauscher (BR) reaction serves as a highly sensitive probe for evaluating free-radical scavenging activity. In this analysis, ferulic acid (FA) was introduced into an actively oscillating BR system to assess its inhibitory effects on the reaction kinetics [2]. Initially, we examined the system’s response under the assumption that FA interacts exclusively with hydroper-

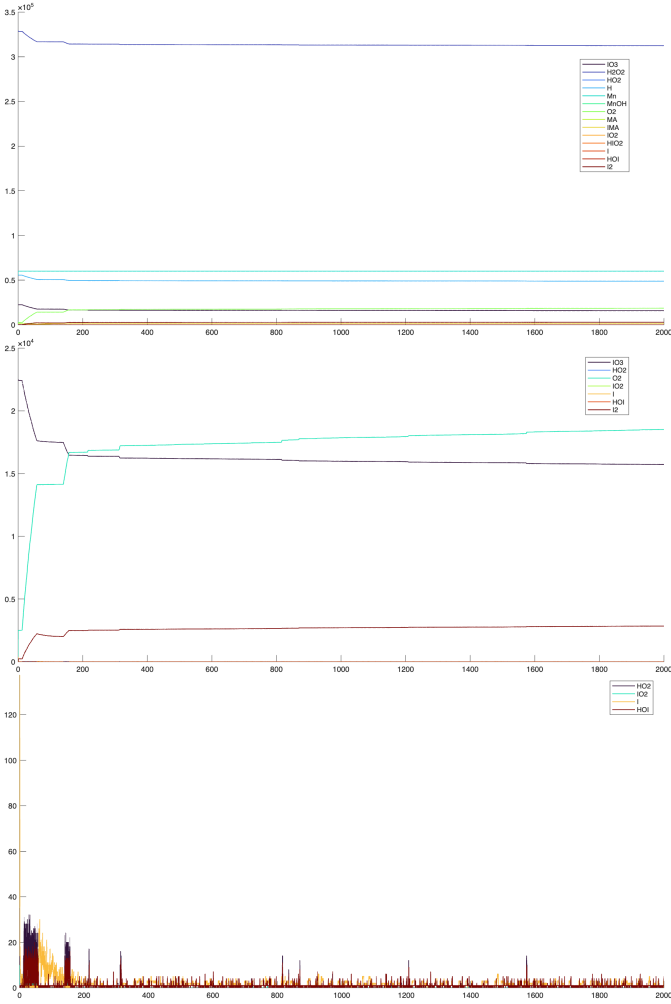
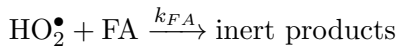


Figure 9

oxyl radicals ($\text{HO}_2^\bullet$). To model this behavior, the following scavenging reaction was added to the Kepper-Epstein model reported in 1:



For the numerical simulation, the following terms were integrated into the system of ordinary differential equations (ODEs) used in section 4:

$$\frac{d[\text{HO}_2^\bullet]}{dt} = k_7 b([\text{Mn}]_0 - [\text{Mn}^{2+}]) - 2k_8 [\text{HO}_2^\bullet]^2 - k_{\text{FA}} [\text{HO}_2^\bullet][\text{FA}]$$

$$\frac{d[\text{FA}]}{dt} = -k_{\text{FA}} [\text{HO}_2^\bullet][\text{FA}]$$

As illustrated in Figure 10, the addition of FA produces a measurable perturbation in the concentration of $\text{HO}_2^\bullet$, consistent with its role as a radical scavenger [2].

However, this perturbation remains largely confined to the $\text{HO}_2^\bullet$ pool and does not propagate to other key intermediates. Consequently, the overall oscillatory dynamics are preserved, and the system resumes oscillations at the characteristic timescale dictated by

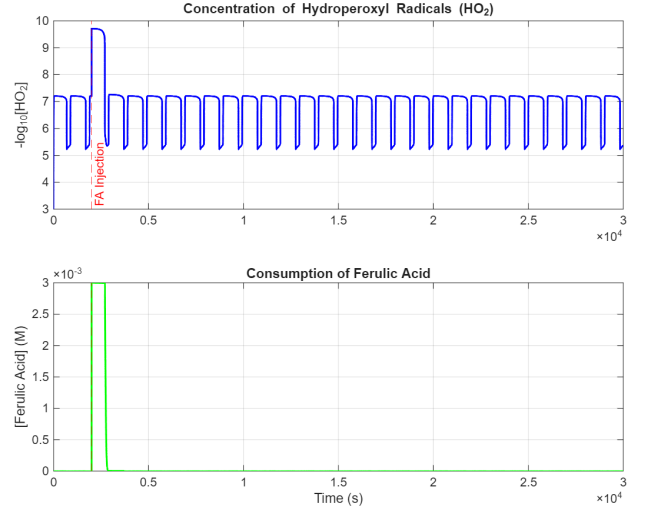


Figure 10: Detailed view of radical suppression and antioxidant consumption in the $\text{HO}_2^\bullet$ model. The top panel shows the transient suppression of $\text{HO}_2^\bullet$ (logarithmic scale). The bottom panel displays the square-wave-like depletion of FA; ($k_{\text{FA}} = 10^4$), indicating near-instantaneous consumption once internal auto-catalytic production exceeds the inhibitory capacity

the intrinsic kinetics of the BR mechanism as shown in Figure 11. This indicates that, under these specific conditions, ferulic acid does not induce a sustained inhibition of the catalytic feedback loops.

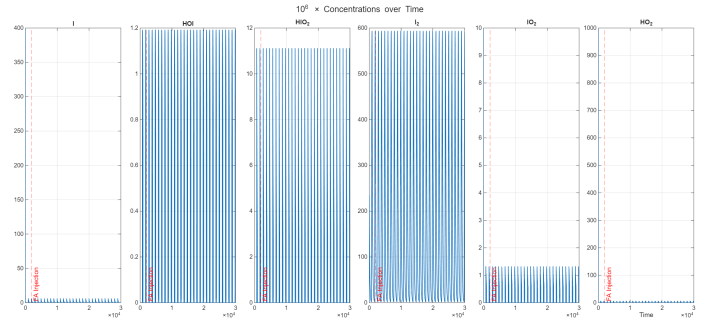
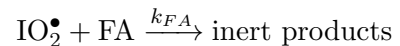


Figure 11: Numerical simulation considering only scavenging of $\text{HO}_2^\bullet$. The dashed red line marks the injection of 3×10^{-3} M FA. ($k_{\text{FA}} = 10^4$)

Following these findings, we simulated a scenario where FA also interacts with the iodine dioxide radical ($\text{IO}_2^\bullet$). For this simulation, we assumed a comparable scavenging efficiency for both radical species and utilized the same rate constant (i.e. $k'_{\text{FA}} = k_{\text{FA}} = 10^4 \text{ M}^{-1} \text{ s}^{-1}$) for both reactions. The following reaction was added to the previous network:



The modified ODEs accounting for the simultaneous consumption of the antioxidant by both radical pools are as follows:

$$\frac{d[\text{IO}_2^\bullet]}{dt} = 2k_4 a h[\text{HIO}_2] - 2k_{12} c[\text{IO}_2^\bullet]^2 - k_6 c[\text{Mn}^{2+}][\text{IO}_2^\bullet] - k'_{\text{FA}} [\text{IO}_2^\bullet][\text{FA}]$$

$$\frac{d[\text{FA}]}{dt} = -k_{\text{FA}} [\text{HO}_2^\bullet][\text{FA}] - k'_{\text{FA}} [\text{IO}_2^\bullet][\text{FA}]$$

In this case, the interaction with both $\text{HO}_2^\bullet$ and $\text{IO}_2^\bullet$ results in a complete suppression of oscillations for several thousand seconds, as shown in Figure 12 and more in detail in Figure 13.

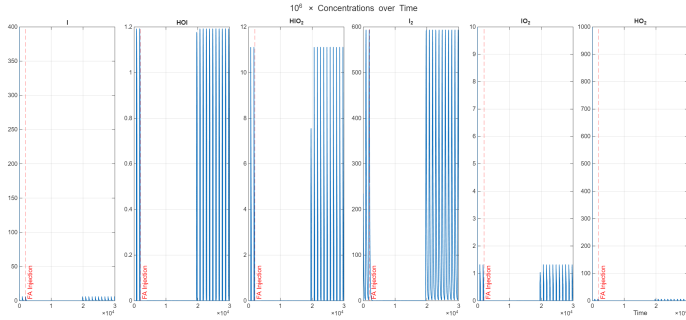


Figure 12: Numerical simulation considering simultaneous scavenging of $\text{HO}_2^\bullet$ and $\text{IO}_2^\bullet$. The dashed red line marks the injection of 3×10^{-3} M FA. ($k_{FA} = 10^4$). It is important to notice the total cessation of oscillations in all chemical species during the inhibition phase.

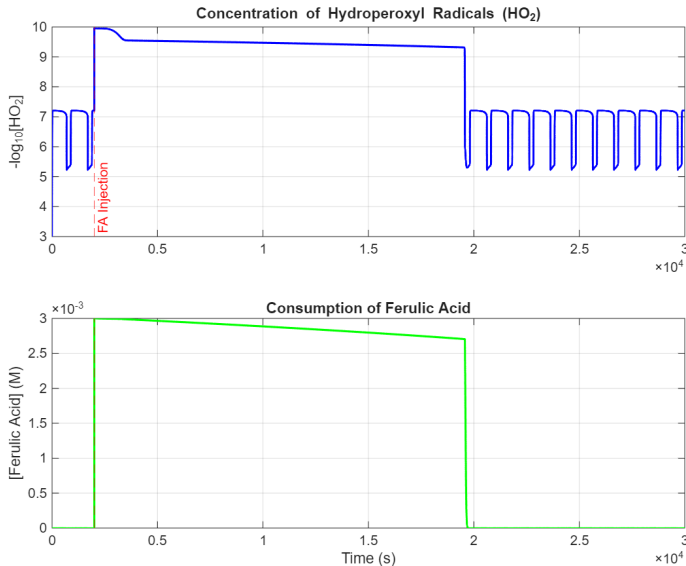


Figure 13: Detailed view of radical suppression and antioxidant consumption in the dual-scavenging model. The top panel illustrates the prolonged suppression of $\text{HO}_2^\bullet$ (logarithmic scale). The bottom panel shows the steady consumption of FA until a critical threshold is reached, triggering the depletion of the remaining scavenger and the resumption of the oscillatory regime.

This inhibition period represents the time required for the system to deplete the antioxidant; oscillations only resume once the concentration of FA falls below a critical threshold, allowing the autocatalytic production of $\text{HO}_2^\bullet$ and $\text{IO}_2^\bullet$ to recover.

These results highlight the inherent limitations of reduced models such as BROCODE, that do not take into account the variability of the radicals. Because such models typically omit the $\text{HO}_2^\bullet$ and $\text{IO}_2^\bullet$ species to achieve mathematical simplicity, they are intrinsically unable to capture the inhibitory dynamics and radical-scavenging interactions necessary to describe antioxidant activity accurately in the Briggs–Rauscher system.

6. Conclusion

This study provides a comprehensive review and critical analysis of the BROCODE model for the Briggs–Rauscher (BR) reaction [1]. While the reduction of the system from fifteen chemical species to three ordinary differential equations successfully preserves the core oscillatory behaviour and greatly enhances analytical transparency, our numerical investigations reveal a pronounced sensitivity to parameter selection.

In particular, the modification of rate constants proposed by the authors to fit the reduced model appears to weaken the assumptions underlying the Quasi-Steady State Approximation (QSSA) for intermediate species such as HOI and $\text{IO}_2^\bullet$. Over specific time intervals, the concentrations of these intermediates become comparable to, or even exceed, those of species that are not treated as quasi-steady, indicating that the reduced system may operate outside its strict domain of mathematical validity despite reproducing the correct qualitative dynamics.

The stochastic formulation highlights intrinsic limitations of the modeling framework. In particular, simulations reveal resolution constraints at small reaction volumes, as the deterministic model predicts fractional molecule numbers while the stochastic formulation enforces integer-valued populations, leading to discrepancies when predicted concentrations fall below one molecule. Species whose deterministic concentrations correspond to less than one molecule may still exhibit non-zero abundances in stochastic simulations. As the reaction volume increases, discrete resolution improves and the stochastic dynamics progressively converge toward the deterministic ODE description, restoring quantitative agreement between the two approaches. Although qualitative oscillatory behavior remains robust across both deterministic and stochastic frameworks, quantitative agreement with the ODE solution is only achieved in the large-volume limit at the cost of substantially increased computational effort, with stochastic simulations requiring hours of runtime compared to seconds for deterministic solvers.

Benchmarking of exact stochastic simulation algorithms further indicates that the Enhanced Direct Method (EDM) consistently outperforms alternatives such as RSSA and FRM, making it the preferred approach for future stochastic investigations of this reaction network.

Finally, this work explores the use of the BR reaction as an antioxidant assay. Our analysis reveals that the suppression of oscillations, commonly referred to as the *inhibition time*, cannot be explained by the scavenging of a single radical species. Instead, inhibition emerges only when the simultaneous scavenging of both $\text{IO}_2^\bullet$ and $\text{HO}_2^\bullet$ radicals is taken into account. This result underscores the chemical complexity of the Briggs–Rauscher system and highlights an

important limitation of reduced models: by eliminating radical intermediates for mathematical simplicity, they become intrinsically incapable of capturing key mechanistic features required to describe antioxidant activity.

In conclusion, while reduced models such as BROCODE are invaluable for analytical insight and qualitative understanding, their predictive reliability depends critically on parameter choices and chemical assumptions whose strength is not robust across parameters selection. Care must therefore be taken when extending such models beyond their intended scope, particularly in applications where radical-level mechanisms play a central role.

Appendix A

6.1. The Next Reaction Method

As discussed in 5.1, the numerical experiments revealed several difficulties when applying the *Next Reaction Method* (NRM) to the Briggs–Rauscher reaction network. For completeness, we briefly recall the algorithmic procedure [6].

Algorithm 1 Next Reaction Method

Input: a biochemical reaction network of M reactions in which each reaction R_j , $j = 1, \dots, M$, is accompanied with the state change vector $\mathbf{v}_j$ and the propensity a_j , the initial state $\mathbf{x}_0$ at time 0 and the simulation ending time T_{max}

Output: a trajectory $X(t)$, $0 \leq t \leq T_{max}$, of the biochemical reaction network

- 1: initialize time $t = 0$ and state $X = \mathbf{x}_0$
 - 2: build the reaction dependency graph G
 - 3: **for all** (reaction R_j) **do**
 - 4: compute a_j
 - 5: generate a random numbers $r_j \sim U(0, 1)$ (see Appendix B.1)
 - 6: set $t_j = (1/a_j) \ln(1/r_j)$
 - 7: **end for**
 - 8: build the binary heap H for M tentative times t_j , $j = 1, \dots, M$
 - 9: **while** ($t < T_{max}$) **do**
 - 10: extract the node with smallest time t_μ and reaction R_μ from heap H
 - 11: set $t = t_\mu$
 - 12: update state $X = X + \mathbf{v}_j$
 - 13: **for all** ($R_j \in \text{Dependents}(R_\mu)$) **do**
 - 14: compute a_j^{new}
 - 15: **if** ($j \neq \mu$) **then**
 - 16: compute $t_j = (a_j/a_j^{new})(t_j - t) + t$
 - 17: **end if**
 - 18: **if** ($j = \mu$) **then**
 - 19: generate a random number $r \sim U(0, 1)$
 - 20: update $t_\mu = t + (1/a_\mu^{new}) \ln(1/r)$
 - 21: **end if**
 - 22: set $a_j = a_j^{new}$
 - 23: replace the old time t_j in H with the new value t_j^{new} and maintain the heap H
 - 24: **end for**
 - 25: **end while**
-

The critical vulnerability lies in the update formula (Line 13). The rescaling of putative times relies on the ratio a_j^{old}/a_j^{new} . This operation becomes ill-defined in two scenarios common to the Briggs–Rauscher dynamics:

1. **Extinction** ($a^{new} \rightarrow 0$): If a reactant is depleted, the new propensity becomes zero. Division by zero yields $t_j = \infty$. While physically consistent

(the reaction cannot fire), this requires the heap structure to handle infinite values robustly.

2. **Propagation of Extinction** ($0 \rightarrow 0$): If a reaction was already impossible ($a^{old} = 0$) and remains impossible ($a^{new} = 0$) after a dependency update, the formula yields the indeterminacy $0/0$, resulting in NaN (Not a Number).

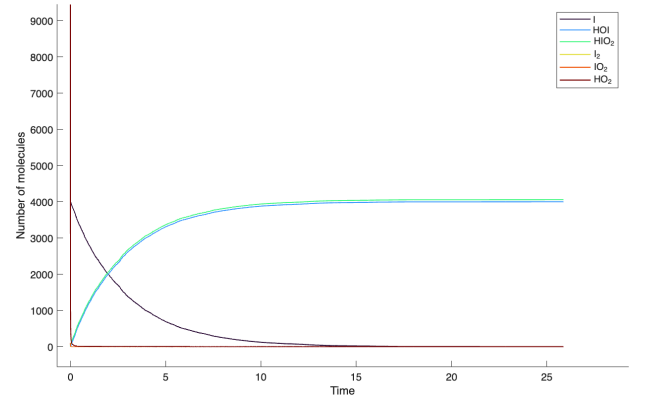


Figure 14: Simulation with NRM, till $T_{max} = 200$ s.

Figure 14 shows a representative trajectory produced by the NRM. Although the simulation was set to run until $T_{max} = 200$ s, reaction events cease at approximately $t \approx 25$ s. This behavior is explained by the occurrence of a reaction with infinite firing time. In practice, the implementation used during the lectures was slightly modified so as to terminate the simulation once such a condition is detected.

MATLAB Command Window

```
>> [T,Dynamics] = simNRM(vMinusBRr,vPlusBRr,cBRr,
    ,initialStateBRr,200,1e-3);

Infinite firing time reached. Stopping
simulation. Current time = 25.8872,
reaction to be fired = 1
Total number of computed simulation steps: 9058
Total number of used random numbers: 9070
Elapsed time is 0.015566 seconds.
```

The reason why this happens, it that propensities are zero for multiple steps in a row, leading to a $0/0$ division, that results in NaN according to Matlab algebra.

MATLAB Command Window

```
>> [T,Dynamics]=simNRM(vMinusBRr,vPlusBRr,
    cBRr,initialStateBRr,200,1e-3);
WARNING: The tentative firing time for reaction
    2 is NaN after firing reaction 8, at step
    1, current time 2.4677e-08.
Old propensity of that reaction: a(2) = 0,
    aNew = 0
Old Propensities a(j):
    0

New Propensities aNew:
    0

Index of broken loop: 2
Total number of computed simulation steps: 0
Total number of used random numbers: 12
Elapsed time is 0.007231 seconds.
```

MATLAB Command Window

```
>> [T,Dynamics]=simNRM(vMinusBRr,vPlusBRr,cBRr,
    initialStateBRr,200,1e-3);
WARNING: Propensity 12 becomes zero after firing
    of reaction 6, at step 73, current time
    0.074618
WARNING: After firing of reaction 6, at step 73,
    current time 0.074618, the tentative
    firing time for reaction 12 is infinite.
WARNING: Propensity 6 becomes zero after firing
    of reaction 6, at step 88, current time
    0.090319
WARNING: after firing reaction 6 at step 88, at
    current time 0.090319, the next tentative
    time tau of that reaction is infinite.
WARNING: Propensity 1 becomes zero after firing
    of reaction 1, at step 7701, current time
    27.033
WARNING: after firing reaction 1 at step 7701,
    at current time 27.033, the next tentative
    time tau of that reaction is infinite.
WARNING: Propensity 2 becomes zero after firing
    of reaction 1, at step 7701, current time
    27.033
WARNING: Propensity 3 becomes zero after firing
    of reaction 1, at step 7701, current time
    27.033
WARNING: Propensity 8 becomes zero after firing
    of reaction 8, at step 7702, current time
    62.7957
WARNING: after firing reaction 8 at step 7702,
    at current time 62.7957, the next tentative
    time tau of that reaction is infinite.
Infinite firing time reached. Stopping
simulation. Current time = 62.7957,
reaction to be fired = 8
Current tau values: NaN NaN NaN NaN NaN NaN NaN
    Inf NaN NaN NaN NaN
Total number of computed simulation steps: 90577
Total number of used random numbers: 90589
Elapsed time is 0.149563 seconds.
```

and "propagate". In the simulation considered here, this situation arises already at the very first steps. The origin of this issue can be formally verified by inspecting the propensities computed using the Direct Method (DM). The issue is that we arrive at a point where the only tentative firing time which is not NaN is infinite. This behavior could have been anticipated, as several chemical species reach and maintain zero abundance during the evolution of the system. To conclude, this example highlights that, beyond formal correctness, the practical applicability of stochastic simulation algorithms crucially depends on the numerical range of the propensities involved.

6.2. The First reaction method

Theoretical singularities associated with vanishing propensities ($a_j \rightarrow 0$) also affect the First Reaction Method (FRM). Formally, as the propensity of a depleted reaction approaches zero, its putative firing time diverges to infinity ($\tau_j \rightarrow \infty$). Formally, the method cannot be applied, as some propensities become null. Even in the case of very small propensities, it can happen to have infinite tentative times. The reason why the method in this case still works, is that the times are re-computed at every step, and it never happens to be left with only infinite times, and the simulation goes on. Although infinite firing times are physically unrealizable, they pose no computational hindrance. As long as the set of putative times contains at least one finite value, the minimization step yields a valid next reaction time, allowing the simulation to proceed without instability. Note also that the First Reaction Method takes in general longer than the DM, since it has to generate more random numbers. Indeed, it requires $M = 12$ random numbers per each time step, against the two required by DM.

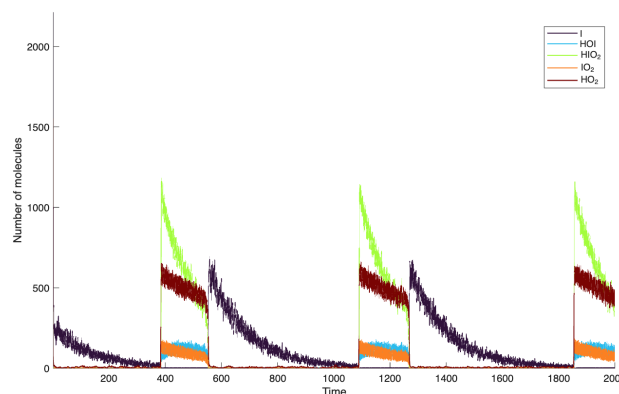


Figure 15: A trajectory simulated with FRM and $V = 10^8/N_A L$, equivalent to those in 4b

Our analysis suggests that after some time we are left with all the tentative times being NaN except one. The NaN originate because of the null propensities,

Appendix B

6.3. Performance analysis: DM and EDM

This section analyzes the runtime differences between the Direct Method (DM) and the Enhanced Direct Method (EDM), linking empirical results to theoretical algorithmic complexities. Table 9 compares the CPU execution times across increasing simulation intervals T_{max} , including an extended run at $T_{max} = 2000$ s (averaged over 100 independent simulations, executed in parallel). Here $V = 10^8/N_{Al}$.

Alg. \ T_{max} [s]	10^2	2×10^2	5×10^2	10^3
EDM	2.4416	3.5238	9.6201	30.0018
DM	6.1134	9.2916	24.3305	74.7792
t_{EDM}/t_{DM}	0.3994	0.3792	0.3954	0.4012

Table 9: Comparison of execution times

Theoretically, the computational cost per step for the standard DM is $\mathcal{O}(M)$ ($M = 12$), as it requires updating the entire propensity vector. Conversely, the EDM updates only the propensities affected by the fired reaction, scaling as $\mathcal{O}(d_{out})$, where d_{out} is the out-degree of the fired reaction in the dependency graph. Based on the empirical ratios of runtimes, the EDM performs as if the effective dependency were $D_{eff} \approx 0.4 \times 12 \approx 4.8$. We may examine the structural dependency graph G :

$$G = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

The arithmetic mean of the out-degrees (row sums) is $\bar{d} \approx 4.8333$. This implies a theoretical speedup ratio of $4.8333/12 \approx 0.402775$. Theory and practice agree almost perfectly.

6.4. Performance Analysis: RSSA vs. RSSA*

In this section, we evaluate the computational efficiency of the Rejection Stochastic Simulation Algorithm (RSSA) and its optimized variant incorporating the dependency graph (RSSA*). A critical observation regarding the applicability of rejection-based methods to the Briggs-Rauscher system must be made. A naive implementation of RSSA demonstrated suboptimal performance for this specific oscillatory dynamics. This inefficiency arises from the frequent depletion of intermediate species: when molecule counts drop to

near-zero levels ($n_i < 10$), the calculation of valid fluctuation intervals may become *ill-posed*. Specifically, low populations lead to degenerate or very small fluctuation bounds, forcing the algorithm to reject a high number of updates or to reduce the time step excessively, thereby negating the theoretical speedup of the approximation. While adaptive heuristics could potentially mitigate this issue, they fall outside the scope of this implementation. This theoretical limitation is clearly evidenced by the simulation results. Specifically, the low populations lead to degenerate fluctuation bounds, forcing the algorithm to update the fluctuation interval in 97.24 % of the steps (59,626,813 updates out of 61,320,736). This statistic confirms that the calculated intervals are often valid for only a single reaction event before the system exits the bound, negating the typical advantage of the algorithm. Furthermore, the algorithm incurred a rejection rate of 9.37 (over 6.3 million rejected steps).

MATLAB Command Window

```
>> [T,Dynamics] = simRSSA*(vMinusBRR,vPlusBRR,
    cBRR,initialStateBRR,0.1,2000,1e-2);

Total number of computed simulation steps:
    60905232
Total number of used random numbers: 201868414
Number of fluctuation interval updates: 59222479
    (97.2371% of steps)
Number of fast acceptance steps: 54683822
    (81.3479%)
Number of slow acceptance steps: 6221410 (9.255%
    )
Number of rejection steps: 6316950 (9.3971%)
Elapsed time is 470.362594 seconds.
```

MATLAB Command Window

```
>> [T,Dynamics] = simRSSA(vMinusBRR,vPlusBRR,
    cBRR,initialStateBRR,0.1,2000,1e-2);

Total number of computed simulation steps:
    61320736
Total number of used random numbers: 203187534
Number of fluctuation interval updates: 59626813
    (97.2376%)
Number of fast acceptance steps: 55046809
    (81.3562%)
Number of slow acceptance steps: 6273927 (9.2725
    %)
Number of rejection steps: 6340713 (9.3712%)
Elapsed time is 1606.059605 seconds.
```

For those interested, we assessed the average execution time of the standard RSSA over a set of 100 independent realizations.

Alg. \ $T_{\max}$ [s]	10^2	2×10^2	5×10^2
RSSA*	9.6303	13.8283	31.8367
RSSA	99.5164	138.5703	397.1434
t_{RSSA^*}/t_{RSSA}	0.0968	0.0997	0.0802

Table 10: Execution times for different methods ($dT = 10^{-2}$), averaged over 100 simulations. * indicates a dependency graph has been exploited. For RSSA, δ was set to 0.1.

Appendix C

This section justifies the choice of the numerical solver employed for the system of 6 ODEs. Analyzing the model’s rate constants reveals that they span a wide range of values. Specifically, there is a difference of 14 orders of magnitude between the slowest and the fastest rates ($k_{-3} = 1.3 \cdot 10^{-2}$ and $k_3 = 3.1 \cdot 10^{12}$, respectively). Consequently, the system exhibits dynamics with widely varying time scales, characterizing it as a stiff system.

Standard explicit methods such as *ode45* (the MATLAB implementation of the *RK45*) are unsuitable for this problem. To maintain stability, the solver is forced to reduce the time step to infinitesimal values to capture the dynamics of the fast reaction (10^{12}). This hypothesis was confirmed by numerical simulations: when using *ode45* with $T_{\max} = 2000$ s, the simulation stalled due to excessive computational cost. In contrast, the stiff solver *ode15s* completed the simulation in just 1.8752 s.

To further illustrate the inefficiency of the explicit method, we compared *ode15s* (full run) against *ode45* run for a limited time ($T_{\max} = 0.0001$ s) to avoid freezing. As shown in Figure 16, the poor performance of *ode45* is directly attributable to the vanishingly small step sizes required by the algorithm.

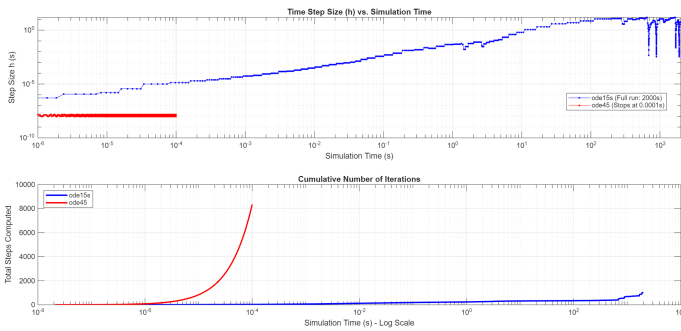


Figure 16: The top panel displays the integration time step size (h) as a function of simulation time (logarithmic scale). The blue trace represents *ode15s*. The red trace shows *ode45*, whose simulation was halted at $t = 10^{-4}$ s due to excessive execution time. The bottom panel illustrates the cumulative number of computed steps. The vertical growth of the *ode45* curve indicates a computational explosion, confirming the stiffness of the system and the unsuitability of explicit methods for this problem.

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